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Surname:															
First Name(s):															

Department of Mathematics and Applied Mathematics
MAM2000W: 2LA (Linear Algebra)

June exam	28 May 2019
Time : 2 hours	Maximum marks: 68
	Full marks: 67

- This question paper consists of 12 pages (including this one).
- Answer all questions in the spaces provided on this question paper. Use the backs of pages when you run out of space.
- Use your *UCT Examination Answer Book* for rough work. We won't take this work in! Only the answers that you write on this question paper will be marked.
- Calculators are not allowed. When your answer contains constants such as π , e or $\sqrt{2}$, leave them in that form. Don't simplify your answers.
- Be careful to provide answers that we can read and make sense of at all times. We will pay attention to your presentation as well as the content. Work that is poorly presented will be penalized. If we can't read your handwriting, we will mark your solution incorrect.

DO NOT WRITE BELOW THIS LINE

Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	TOTAL
17	4	6	10	11	10	5	5	68

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Question 1. [17 points] Let $A = \begin{pmatrix} 3 & 2 & 1 \\ 0 & 5 & 0 \\ 6 & 1 & 2 \end{pmatrix}$.

Write the answers to the following questions in the blocks provided. **Do not show working!!!**

a. Find $\det(A)$. (1)

$$\det(A) = 0.$$

b. Is it possible to write A as a product of elementary matrices? Give a reason for your answer that does not require any further calculation. (1)

No, since $\det(A) = 0$ (so A is not invertible).

c. Find elementary matrices E_1 and E_2 and an upper triangular matrix U such that $A = E_2E_1U$. (3)

$$E_1 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -\frac{3}{5} & 1 \end{pmatrix}, \quad E_2 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 2 & 0 & 1 \end{pmatrix}, \quad U = \begin{pmatrix} 3 & 2 & 1 \\ 0 & 5 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

(1 Mark for each matrix, but if they forget to invert the elementary matrices just take off one mark.)

d. Find a basis for the nullspace of A . (1)

$$\left\{ \begin{pmatrix} 1 \\ 0 \\ -3 \end{pmatrix} \right\}$$

e. Find a basis for the column space of A . (1)

$$\left\{ \begin{pmatrix} 3 \\ 0 \\ 6 \end{pmatrix}, \begin{pmatrix} 2 \\ 5 \\ 1 \end{pmatrix} \right\}$$

f. Find a basis for the row space of A . (1)

$$\{(3 \ 2 \ 1), (0 \ 5 \ 0)\}$$

g. What is the rank and nullity of A ? (1)

Nullity of $A = 1$, Rank of $A = 2$.

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- h. Find all the eigenvalues of A , together with their multiplicities.

(2)

$\lambda = 5$ with multiplicity 2, $\lambda = 0$ with multiplicity 1.

- i. Can you say, without further calculation, whether A is diagonalizable? Give a reason for your answer.

(1)

No. There are only two distinct eigenvalues (they can say 5 has multiplicity 2) and therefore it is not possible to say whether there will be 3 linearly independent eigenvectors without first calculating eigenvectors.

- j. For each eigenvalue of A , find the corresponding eigenspace.

(4)

$$W_0 = \left\{ t \begin{pmatrix} -1 \\ 0 \\ 3 \end{pmatrix} : t \in \mathbb{R} \right\}, \quad W_5 = \left\{ t \begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix} : t \in \mathbb{R} \right\}.$$

- k. Is A diagonalizable? Give a reason for your answer.

(1)

No, there are only 2 linearly independent eigenvectors, not 3 as needed.

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Question 2. [4 points] Let A be an $m \times n$ matrix. Complete each of the following statements by choosing one of the options (A) - (G) below. You may use an option more than once, and you do not have to use all options.

a. If $A\mathbf{x} = \mathbf{b}$ is consistent for all $\mathbf{b} \in \mathbb{R}^m$, then

A or D or G.

b. If A is non-singular, then

B or G.

c. If A is singular, then

E.

d. If $m = n$ and A is invertible, then

G or A or B or C or D?

(A) the rows of A are linearly independent.

(B) the columns of A are linearly independent.

(C) the rows of A span $\mathbb{R}^n$.

(D) the columns of A span $\mathbb{R}^m$.

(E) $\text{Nullity}(A) > 0$.

(F) A is diagonalizable.

(G) 0 is not an eigenvalue of A .

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Question 3. [6 points] Complete the following:

- a. If A is a 2×2 matrix with determinant 6 and trace 5, then its eigenvalues are... (1)

2 and 3.

- b. Let $T : P_3 \rightarrow P_3$ be a linear transformation for which

$$T(x^3) = x^2, \quad T(x^2) = x^2, \quad T(x) = x, \quad T(1) = x.$$

Then $T(5x^2 + 3) = \boxed{}$ and $T(0) = \boxed{}$ (2)

$T(5x^2 + 3) = 5x^2 + 3x$ and $T(0) = 0$.

- c. Let $B = \begin{pmatrix} 7 & 4 \\ -8 & -5 \end{pmatrix}$. You are given that

$$B \begin{pmatrix} 1 \\ -1 \end{pmatrix} = 3 \begin{pmatrix} 1 \\ -1 \end{pmatrix} \quad \text{and} \quad B \begin{pmatrix} 1 \\ -2 \end{pmatrix} = - \begin{pmatrix} 1 \\ -2 \end{pmatrix}.$$

Find an invertible matrix P which diagonalizes B , and the resulting diagonal matrix D . Write down an expression relating the matrices B , P and D . (3)

$P = \begin{pmatrix} 1 & 1 \\ -1 & -2 \end{pmatrix}$, and $D = \begin{pmatrix} 3 & 0 \\ 0 & -1 \end{pmatrix}$. $P^{-1}BP = D$ or $BP = PD$.

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Question 4. [10 points]

In P_3 , let $S = \{x^3 + x, x^2 + 2x - 1, x, x^2 - 1\}$.

Prove or disprove each of the following statements. **Show all your working!**

a. S spans P_3 .

(4)

Consider the equation

$$\lambda_1(x^3 + x) + \lambda_2(x^2 + 2x - 1) + \lambda_3x + \lambda_4(x^2 - 1) = ax^3 + bx^2 + cx + d.$$

(To get this mark they need to set up the problem – either like this, or by saying something like “ $ax^3 + bx^2 + cx + d$ is in $\text{span}(S)$ if and only if there is a solution to the following system.”)

This leads to the system of equations:

$$\begin{aligned}\lambda_1 &= a \\ \lambda_2 + \lambda_4 &= b \\ \lambda_1 + 2\lambda_2 + \lambda_3 &= c \\ -\lambda_2 - \lambda_4 &= d\end{aligned}$$

with augmented matrix

$$\begin{pmatrix} 1 & 0 & 0 & 0 & a \\ 0 & 1 & 0 & 1 & b \\ 1 & 2 & 1 & 0 & c \\ 0 & -1 & 0 & -1 & d \end{pmatrix}$$

(one mark for this)

which on Gauss reduction yields

$$\begin{pmatrix} 1 & 0 & 0 & 0 & a \\ 0 & 1 & 0 & 1 & b \\ 0 & 0 & 1 & -2 & c - a - 2b \\ 0 & 0 & 0 & 0 & d + b \end{pmatrix}.$$

(One mark for the solution – they don’t actually need to go all the way to the echelon form though, since they’re not trying to find the span, just show it’s not the whole vector space!)

This has a solution if and only if $d = -b$, so S does not span P_3 . (To get this mark their conclusion should be based on their echelon matrix, of course. If it has no zero rows then they should conclude that S does span P_3 .)

OR could prove S is linearly dependent, so we can drop some vector from S without changing its span. But this new set has 3 vectors, while the dimension of P_3 is 4, so it cannot span P_3 .

b. S is linearly independent

(2)

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S is not linearly independent, since

$$x^2 + 2x - 1 - 2(x) - (x^2 - 1) = 0.$$

OR Because the row-echelon form of the augmented matrix has one free variable, and hence the homogeneous system obtained by taking $a = b = c = d = 0$ will have non-trivial solutions.

OR Because S has $4 = \dim(P_3)$ vectors in it, so if it were linearly dependent then it would span P_3 , and we have proved that it does not.

c. S is a basis for P_3 . (1)

S is not a basis, since it is not linearly independent. OR Since it does not span P_3 .

d. $\dim(\text{span}(S)) = 3$. (3)

$\dim(\text{span}(S)) = 3$, since

1.

$$x^2 + 2x - 1 - 2(x) - (x^2 - 1) = 0,$$

so any one of the last three vectors in S can be dropped without changing the span, and

2. the three remaining vectors are linearly independent, therefore form a basis for the span (Proof: if $a = b = c = d = 0$ then $\lambda_2 = 0$ or $\lambda_3 = 0$ or $\lambda_4 = 0$ implies $\lambda_1 = \lambda_3 = \lambda_4 = 0$) and so $\dim(\text{span}(S')) = 3$.

OR Leaving out the second, third or fourth column of the echelon matrix leaves it with column rank 3 / no free variables / all columns linearly independent, therefore the only solution for the remaining λ_i is the trivial one.

OR

1. S is not a basis (hence $\dim(\text{span}(S)) < 4$) and

2. Prove three elements of S are linearly independent (as above), and so $\dim(\text{span}(S)) \geq 3$.

OR

$$\text{span}(S) = \{ax^3 + b(x^2 - 1) + cx \mid a, b, c \in \mathbb{R}\},$$

where x^3 , $x^2 - 1$ and x are linearly independent (needs proof) and therefore form a basis for the span. So dimension is 3.

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Question 5. [11 points]

For each of the following statements, write **True** or **False**, and justify your answer. (No marks will be given for just writing “True” or “False” with no explanation, or with an incorrect explanation.)

- a. Suppose that after Gauss reducing the augmented coefficient matrix of a system of linear equations, we get the following matrix:

$$\left(\begin{array}{ccc|c} 1 & -2 & 0 & 4 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right)$$

The solution of this system is a plane in $\mathbb{R}^3$. (2)

False. The solution is a line in $\mathbb{R}^3$. (It's $\{(x, y, z) \mid x = 4 + 2y, y \in \mathbb{R}, z = 0\}$, which is a line in the xy plane, but you didn't have to say that.)

- b. If B is a non-zero matrix and $B = E_k \cdots E_1 U$ where $E_1, \dots, E_k$ are elementary matrices and U is in row echelon form, then B is invertible. (2)

False. If U has a row of zeroes, then B is not invertible.

- c. The set of all $A \in M_{2 \times 2}(\mathbb{R})$ such that $A^2 = \mathbf{0}$ is closed under scalar multiplication. (2)

True. If $A \in M_{2 \times 2}(\mathbb{R})$ and $\lambda \in \mathbb{R}$, then $(\lambda A)^2 = \lambda^2 A^2 = \lambda^2 \mathbf{0} = \mathbf{0}$.

- d. The set of all $A \in M_{2 \times 2}(\mathbb{R})$ such that $A^2 = \mathbf{0}$ is a subspace of the real vector space $M_{2 \times 2}(\mathbb{R})$. (2)

False. $M_{2 \times 2}(\mathbb{R})$ is not closed under addition, since for instance $\begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$.

- e. If A and B are $n \times n$ matrices such that $AB = \mathbf{0}$, then $\text{rank}(A) + \text{rank}(B) \leq n$. (3)

True. Each column of B is in the nullspace of A , so $\text{rank}(B) \leq \text{nullity}(A)$, and hence $\text{rank}(A) + \text{rank}(B) \leq \text{rank}(A) + \text{nullity}(A) = n$.

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Question 6. [10 points] Read the theorem and its proof below and then answer the questions that follow.

Theorem If $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_m$ are eigenvectors of a square matrix A associated with distinct eigenvalues $\lambda_1, \lambda_2, \dots, \lambda_m$ respectively, then $S = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_m\}$ is linearly independent.

Proof: Suppose that S is not linearly independent. We certainly know that $\{\mathbf{v}_1\}$ is linearly independent, so let $k > 1$ be the first integer such that $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k\}$ is linearly dependent. Then there are scalars $a_1, a_2, \dots, a_k$, not all zero, such that

$$a_1 \mathbf{v}_1 + a_2 \mathbf{v}_2 + \dots + a_k \mathbf{v}_k = \mathbf{0}. \quad (1)$$

If we multiply both sides of (1) on the left by A , we get

$$a_1 \lambda_1 \mathbf{v}_1 + a_2 \lambda_2 \mathbf{v}_2 + \dots + a_k \lambda_k \mathbf{v}_k = \mathbf{0}. \quad (2)$$

Now multiply (1) by λ_k :

$$a_1 \lambda_k \mathbf{v}_1 + a_2 \lambda_k \mathbf{v}_2 + \dots + a_k \lambda_k \mathbf{v}_k = \mathbf{0}. \quad (3)$$

If we now subtract (2) from (3) we get:

$$a_1(\lambda_k - \lambda_1) \mathbf{v}_1 + a_2(\lambda_k - \lambda_2) \mathbf{v}_2 + \dots + a_{k-1}(\lambda_k - \lambda_{k-1}) \mathbf{v}_{k-1} = \mathbf{0}. \quad (4)$$

From this we can deduce that

$$a_1(\lambda_k - \lambda_1) = a_2(\lambda_k - \lambda_2) = \dots = a_{k-1}(\lambda_k - \lambda_{k-1}) = 0. \quad (5)$$

Therefore we must have $a_1 = a_2 = \dots = a_{k-1} = 0$. (6)

Going back to equation (1) this leaves us with $a_k \mathbf{v}_k = \mathbf{0}$ and from this we can deduce that $a_k = 0$. (7)

But this gives a contradiction, (8)

so the set S must be linearly independent.

a. How do we know $\{\mathbf{v}_1\}$ is linearly independent? (2)

$\mathbf{v}_1$ is an eigenvector, so it is non-zero.

b. Explain carefully how equation (2) follows from equation (1). (2)

By the distributive laws,

$$A(a_1 \mathbf{v}_1 + a_2 \mathbf{v}_2 + \dots + a_k \mathbf{v}_k) = a_1 A \mathbf{v}_1 + a_2 A \mathbf{v}_2 + \dots + a_k A \mathbf{v}_k.$$

Because each $\mathbf{v}_i$ is an eigenvector, $A \mathbf{v}_i = \lambda_i \mathbf{v}_i$ for each i

c. In deducing (5) from (4), which property of the set $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_{k-1}\}$ are we using? How do we know it has this property? (2)

The set is linearly independent by our choice of k .

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d. Why can we deduce at (7) that $a_k = 0$?

(1)

We have that $\mathbf{v}_k \neq \mathbf{0}$ because $\mathbf{v}_k$ is an eigenvector.

e. What is the contradiction referred to at (8)?

(1)

All the scalars $a_1, a_2, \dots, a_k$ are zero, contrary to the initial assumption. OR Contradicts the minimality of k .

f. Where in the proof is the fact that the eigenvalues are distinct used, and how?

(2)

It is used in going from (5) to (6). For each i , we have that $a_i(\lambda_k - \lambda_i) = 0$, so either $a_i = 0$ or $\lambda_k = \lambda_i$. The second of these possibilities is excluded because $\lambda_k \neq \lambda_i$.

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Question 7. [5 points] Prove the following theorem.

Theorem: Let $S = \{\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_n\}$ be a non-empty set of vectors in a vector space V . If S is linearly independent, and if $\mathbf{v}$ is a vector in V that is not in the span of S then $S \cup \{\mathbf{v}\}$ is still linearly independent.

Suppose $S \cup \{\mathbf{v}\}$ is linearly dependent. Then there are scalars $\lambda_1, \lambda_2, \dots, \lambda_{n+1}$, not all zero, such that

$$\lambda_1 \mathbf{u}_1 + \lambda_2 \mathbf{u}_2 + \dots + \lambda_n \mathbf{u}_n + \lambda_{n+1} \mathbf{v} = \mathbf{0}.$$

We cannot have $\lambda_{n+1} = 0$, because S is linearly independent. So $\lambda_{n+1} \neq 0$, and

$$\mathbf{v} = -\frac{1}{\lambda_{n+1}} (\lambda_1 \mathbf{u}_1 + \lambda_2 \mathbf{u}_2 + \dots + \lambda_n \mathbf{u}_n).$$

But this contradicts the fact that $\mathbf{v} \notin \text{span}(S)$, so $S \cup \{\mathbf{v}\}$ must be linearly independent.

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Question 8. [5 points] For a square matrix A , let $\sigma(A)$ denote the set of all eigenvalues of A . Prove that if A and B are square matrices of the same size, then $\sigma(AB) = \sigma(BA)$.

Let $\lambda \in \sigma(AB)$.

If $\lambda = 0$, then AB is not invertible, and then neither is BA , so $\lambda = 0 \in \sigma(BA)$.

If $\lambda \neq 0$, then let $\mathbf{x}$ be any eigenvector of AB corresponding to λ . Then

$$AB\mathbf{x} = \lambda\mathbf{x},$$

so

$$(BA)(B\mathbf{x}) = B(AB)\mathbf{x} = B(\lambda\mathbf{x}) = \lambda(B\mathbf{x}).$$

Note that $B\mathbf{x} \neq \mathbf{0}$, since otherwise $AB\mathbf{x} = \lambda\mathbf{x}$ would be $\mathbf{0}$. So λ is an eigenvalue of BA with eigenvector $B\mathbf{x}$, and $\lambda \in \sigma(BA)$. Thus $\sigma(AB) \subseteq \sigma(BA)$. Similarly, $\sigma(BA) \subseteq \sigma(AB)$. So $\sigma(AB) = \sigma(BA)$.